Venkata Lakshmi Parimala Pasupuleti

Software Developer · Agentic AI Workflows · Clinical & Research Data Pipelines · Reproducible Systems · Cloud Integration · Python · AWS

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SUMMARY

Software developer with 9+ years of experience building **reproducible agentic AI workflows**, research data pipelines, and secure cloud integrations. As a Research Associate at George Mason University, built AI-powered data preprocessing pipelines and anomaly detection tools for WHO and UK Road Safety research datasets - delivering reproducible, validated outputs to faculty researchers. Built **AI on FHIR** - a clinical AI system translating natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings, with HIPAA-style RBAC and OAuth-secured clinical data integrations. Currently at Intuit building **agentic AI workflows** with Anthropic Claude for production financial services: tool calling, retrieval-grounded context, structured outputs, guardrails, and CI evaluation harnesses. Built **Leasel** - an 8-agent LangGraph platform with reproducible CI eval (93.6% precision, 1,240 test cases), and **AgentProbe** - an open-source eval and observability platform with reproducible golden-dataset evals and CI regression gating. At Entrata (7+ years), owned PHI-compliant AWS data systems and compliance ETL pipelines under regulated frameworks. Strong in Python, FastAPI, AWS cloud integrations, secure credential management, Docker, Kubernetes, and collaborative research environments. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Reproducible Agentic AI Workflows · Research Data Pipelines | Clinical AI · FHIR · Biomedical Data · Healthcare Integrations

Cloud Integrations · Secure Credential Management · Identity Controls | Research Software Development · Data Validation · Reproducibility

LLM Agents · LangGraph · Tool Calling · Multi-Step Workflows | LLM Evaluation · CI Regression · Golden Datasets · Observability

Python · R · FastAPI · Flask · Data Engineering | AWS (S3 · Lambda · SQS · IAM) · Docker · Kubernetes · CI/CD

PostgreSQL · SQL · ETL · Async Processing · Event-Driven | Anomaly Detection · Statistical Modeling · scikit-learn · XGBoost

Technical Communication · Researcher Collaboration · Mentorship | PHI-Compliant Systems · HIPAA · Regulated Data Environments

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **Agentic AI workflows in production:** Build LangChain-based autonomous AI workflows with tool calling, retrieval-grounded context management, structured outputs, and cloud integrations; guardrails with input validation and fallbacks; CI evaluation harnesses with LLM-as-judge and golden datasets.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS with structured logging and CloudWatch monitoring.

Research Associate

Aug 2024 - Aug 2025

George Mason University — Fairfax, Virginia

- **Research data pipelines:** Built reproducible data preprocessing pipelines and regression models ($R^2 = 0.977$) in Python/R on WHO and UK Road Safety datasets (190+ countries); delivered validated, reproducible outputs for peer-reviewed research workflows.
- **AI-powered research tooling:** Built AI-powered anomaly detection auditor (Python, Isolation Forest) - reduced manual QA time by 70% across 3 research projects; created Tableau, ggplot2, and Quarto dashboards that improved faculty comprehension of complex datasets by 40%.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing SaaS (regulated multi-tenant, 100K+ users)

- Owned PHI-compliant AWS S3 data migration (Libsodium encryption, secure credential management, 100% coverage, zero incidents); HUD compliance ETL (180K records, zero violations across 6 years); RabbitMQ async event-driven processing with idempotency guarantees.
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

AI on FHIR — Clinical AI & Healthcare API Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language clinical questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. Secure OAuth API integration, HIPAA-style RBAC with identity controls, Docker-containerized deployment. Demonstrates direct clinical data integration experience relevant to biomedical research workflows.

Leasel - Reproducible Agentic AI Platform: LangGraph · CI Evals · Cloud Integration

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic AI platform: 8-agent LangGraph orchestration with tool calling, durable state, and conditional routing; hybrid search RAG (OCR ingestion, vector search, semantic reranking); HITL guardrails for regulated decisions; **reproducible CI eval harness** (LLM-as-judge, golden datasets:

93.6% precision across 1,240 test cases). Cloud-deployed with full observability and structured logging.

AgentProbe - Reproducible LLM Evaluation & Observability Platform 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source platform for **reproducible LLM evaluation**: offline golden-dataset evals, LLM-as-judge scoring, CI regression gating, failure handling; cost-aware model routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing and structured logging.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017